Reading List: From Linear Conjugate Gradient to Modern Least-Squares Solvers

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Read in this order. Each entry says what the paper contributes and what to reproduce from it.

1 Linear conjugate gradient

1. **M. R. Hestenes, E. Stiefel.** Methods of conjugate gradients for solving linear systems. *J. Res. Nat. Bur. Standards* 49(6):409–436, 1952.
   The original CG. Reproduce: conjugacy, finite termination, minimization property. §10 already gives CG for least squares without forming $J^T J$.

2. **C. Lanczos.** Solution of systems of linear equations by minimized iterations. *J. Res. Nat. Bur. Standards* 49(1):33–53, 1952.
   The parallel Lanczos view of the same Krylov process; needed later for Golub–Kahan.

3. **J. K. Reid.** On the method of conjugate gradients for the solution of large sparse systems of linear equations. In *Large Sparse Sets of Linear Equations*, Academic Press, 231–254, 1971.
   CG revived as an *iterative* method (stop well before n steps).

4. **G. H. Golub, D. P. O'Leary.** Some history of the conjugate gradient and Lanczos algorithms: 1948–1976. *SIAM Review* 31(1):50–102, 1989.
   Map of the early literature; use it to place every other paper.

5. **A. Greenbaum.** Behavior of slightly perturbed Lanczos and conjugate-gradient recurrences. *Linear Algebra Appl.* 113:7–63, 1989; and **A. Greenbaum, Z. Strakoš.** Predicting the behavior of finite precision Lanczos and conjugate gradient computations. *SIAM J. Matrix Anal. Appl.* 13(1):121–137, 1992.
   Why CG in floating point loses orthogonality and is delayed but still converges.

6. **G. Meurant, Z. Strakoš.** The Lanczos and conjugate gradient algorithms in finite precision arithmetic. *Acta Numerica* 15:471–542, 2006.
   Survey of the finite-precision theory; reproduce the delay of convergence.

2 CG for linear least squares

7. **G. H. Golub.** Numerical methods for solving linear least squares problems. *Numer. Math.* 7:206–216, 1965.
   Why the normal equations ($\kappa(J)^2$) should be avoided; QR as the reference method.

8. **G. H. Golub, W. Kahan.** Calculating the singular values and pseudo-inverse of a matrix. *SIAM J. Numer. Anal. Ser. B* 2(2):205–224, 1965.
   The bidiagonalization that LSQR and LSMR are built on.

9. **C. C. Paige.** Bidiagonalization of matrices and solution of linear equations. *SIAM J. Numer. Anal.* 11:197–209, 1974.
   First use of Golub–Kahan bidiagonalization to solve systems; precursor of LSQR.

10. **G. W. Stewart.** Research, development, and LINPACK. In *Mathematical Software III*, Academic Press, 1–14, 1977.
    Backward error of an approximate least-squares solution, $\|E\| = \|J^T r\|/\|r\|$; the basis of the LSQR stopping rules.

11. **C. C. Paige, M. A. Saunders.** LSQR: an algorithm for sparse linear equations and sparse least squares. *ACM Trans. Math. Softw.* 8(1):43–71, 1982.
    Central paper. Reproduce: equivalence with CGLS, the Givens-rotation implementation, the estimates of $\|J\|$ and $\kappa(J)$, and the stopping rules.

12. **Å. Björck, T. Elfving, Z. Strakoš.** Stability of conjugate gradient and Lanczos methods for linear least squares problems. *SIAM J. Matrix Anal. Appl.* 19(3):720–736, 1998. doi:10.1137/S089547989631202X.
    Which of the algebraically equivalent CGLS implementations is accurate in floating point, and how CGLS compares with LSQR.

13. **X.-W. Chang, C. C. Paige, D. Titley-Peloquin.** Stopping criteria for the iterative solution of linear least squares problems. *SIAM J. Matrix Anal. Appl.* 31:831–852, 2009.
    Refined backward-error stopping rules.

14. **D. C.-L. Fong, M. A. Saunders.** LSMR: an iterative algorithm for sparse least-squares problems. *SIAM J. Sci. Comput.* 33(5):2950–2971, 2011. arXiv:1006.0758.
    MINRES on the normal equations; $\|J^T r_k\|$ decreases monotonically. Reproduce the comparison with LSQR.

15. **R. Estrin, D. Orban, M. A. Saunders.** LSLQ: an iterative method for linear least-squares with an error minimization property. *SIAM J. Matrix Anal. Appl.* 40(1):254–275, 2019.
    Third member of the family: minimizes the error and gives error estimates.

3 Randomized preconditioning (current research)

16. **V. Rokhlin, M. Tygert.** A fast randomized algorithm for overdetermined linear least-squares regression. *Proc. Natl. Acad. Sci. USA* 105(36):13212–13217, 2008.
    First sketch-and-precondition solver.

17. **H. Avron, P. Maymounkov, S. Toledo.** Blendenpik: supercharging LAPACK's least-squares solver. *SIAM J. Sci. Comput.* 32(3):1217–1236, 2010. doi:10.1137/090767911.
    Practical sketch preconditioner + LSQR; reproduce the condition number of JR^{-1}.

18. **X. Meng, M. A. Saunders, M. W. Mahoney.** LSRN: a parallel iterative solver for strongly over- or underdetermined systems. *SIAM J. Sci. Comput.* 36(2):C95–C118, 2014.
    Gaussian-sketch preconditioner with guaranteed condition number; parallel version.

19. **M. Meier, Y. Nakatsukasa, A. Townsend, M. Webb.** Are sketch-and-precondition least squares solvers numerically stable? *SIAM J. Matrix Anal. Appl.* 45(2):905–929, 2024. arXiv:2302.07202.
    Shows Blendenpik/LSRN with zero initial guess are unstable; recommends the sketch-and-solve start.

20. **E. N. Epperly.** Fast and forward stable randomized algorithms for linear least-squares problems. *SIAM J. Matrix Anal. Appl.* 45:1782–1804, 2024. doi:10.1137/23M1616790.
    Proves forward stability of the corrected method.

21. **E. N. Epperly, M. Meier, Y. Nakatsukasa.** Fast randomized least-squares solvers can be just as accurate and stable as classical direct solvers. *Comm. Pure Appl. Math.* 79(2):293–

339, 2026. arXiv:2406.03468.
Backward stable randomized solvers (SPIR, FOSSILS): the current state of the art.

4 Reference books

- J. Nocedal, S. J. Wright. *Numerical Optimization*, 2nd ed., Springer, 2006, Ch. 5 (linear CG).
- Å. Björck. *Numerical Methods for Least Squares Problems*, SIAM, 1996 (normal equations, QR, CGLS, LSQR).
- N. J. Higham. *Accuracy and Stability of Numerical Algorithms*, 2nd ed., SIAM, 2002, Ch. 20 (least-squares error analysis).
- J. Liesen, Z. Strakoš. *Krylov Subspace Methods: Principles and Analysis*, Oxford University Press, 2013.
- Y. Saad. *Iterative Methods for Sparse Linear Systems*, 2nd ed., SIAM, 2003.